AI-POWERED CYBERSECURITY

Assessment 01
Security Foundations

Final Report – Tasks A, B, C and D

Field	Details
Student name	Sanjay Kushwaha
Batch / group	Ewb AI POWERED CYBERSECURITY
Assessment	Individual work | 100 marks
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This report analyses fictional training scenarios only. No real accounts, passwords, OTPs, or live websites were tested.

Task A – Secure a Student Account (25 Marks)

A1. Five risks, assets, CIA impact, controls and limitations

Risk	Asset at risk	CIA impact	Control	Limitation
Reusing a short password	Email and learning portal	Confidentiality: if exposed, someone may access both accounts.	Use a long, unique password for each account.	Phishing or malware can still steal passwords.
Approving unexpected prompts	Online accounts and private information	Confidentiality and integrity: someone may view or change data.	Reject unexpected prompts; enable MFA and report repeated prompts.	Some MFA methods can still be tricked or bypassed.
Delaying updates	Laptop, apps and files	Confidentiality, integrity, availability: known weaknesses may be exploited.	Enable automatic security updates and install them promptly.	Updates cannot prevent every attack.
Only one project copy	Academic work	Availability: failure, theft or ransomware could make work unavailable.	Keep secure backups in cloud storage or on an external drive; test restoration.	Backups may be old, damaged or compromised.
Friend uses signed-in laptop	Files, email and learning portal	Confidentiality and integrity: private information may be viewed or files changed/deleted.	Create a separate standard user account and lock Mira's session.	Poor permissions or shared credentials may still allow access.

CIA means Confidentiality (only authorised people can see information), Integrity (information is not changed without permission), and Availability (information and systems can be used when needed).

A2. First three priorities

1. Secure account access first. Change the reused password, use unique passwords, enable MFA, and reject unexpected prompts. This comes first because account access could expose private information. Limitation: these controls cannot stop every attack.

2. Protect project files. Make a secure backup and test that it can be restored. This protects work if the laptop breaks, is stolen, or is affected by ransomware. Limitation: a backup may not include the newest work if it is not updated.

3. Secure and update the laptop. Install pending security updates, enable automatic updates, create a separate account for the friend, and lock the device when away. This reduces known weaknesses and unnecessary access. Limitation: these controls do not prevent every incident.

A3. Authentication versus authorisation

Authentication checks who a user is. Mira enters her username and password and completes an MFA check to sign in to the learning portal. Authorisation decides what the signed-in user is allowed to do. Mira may view course materials and submit assignments, but cannot change another student's grades or access restricted administrator settings.

Task B – Analyse a Suspicious Message (25 Marks)

B1. Five warning signs

Warning sign (quote)	Why it needs scrutiny
“FINAL WARNING - scholarship expires in 20 minutes”	Creates pressure to act without checking.
“From: ... awards@campus-awards.example”	The domain differs from the college's usual campus.example domain. This needs checking but is not proof by itself.
“Reply-To: help@personal-awards.example”	The reply address uses a different domain from the college and sender address.
“enter your college password and OTP”	A password and OTP request could let someone access the account.
“Do not contact the college office”	Discourages independent verification through an official channel.

An additional warning sign is the request for a photo of an identity card, because it could expose sensitive personal information.

B2. Safe verification process

1. Do not click the supplied link or enter details. 2. Open the college's known portal using a saved bookmark or an address obtained independently. 3. Contact the scholarship office using details published through an official college source. 4. Report the message to the official IT/security team. 5. Wait for confirmation before sharing information.

HTTPS encrypts the connection between a browser and a website; it does not prove the site is genuine. A familiar display name can also be copied. Neither alone proves the message is legitimate.

B3. If a student already entered a password and OTP

1. From a trusted device, open the official portal directly and change the exposed password; do not reuse it elsewhere. 2. Contact the college IT/security team immediately. 3. Ask IT to revoke active sessions and check for unauthorised access. 4. Change the password on other accounts where it was reused. 5. Review MFA settings with IT guidance and remove unfamiliar devices or methods. 6. Preserve the message, sender details and incident time; report through an official channel. 7. Monitor for unfamiliar logins, account changes or messages. If an identity document was shared, inform the college. Never send passwords or OTPs to support staff.

B4. What cannot be established

The message alone cannot prove who sent it or whether the scholarship offer is false. Helpful additional evidence includes full email headers, sender-authentication results, relevant security logs, and confirmation from official scholarship records.

Task C – Triage a Sign-in Alert (25 Marks)

C1. Four-point timeline (18 October 2026, IST)

Time (IST)	Observed evidence	Interpretation / uncertainty
09:00–09:01	Password success and MFA approval from usual classroom IP 192.0.2.10.	Consistent with a normal sign-in, but does not prove who signed in.
10:14–10:16	Password failures from 198.51.100.23.	May be repeated access attempts; the IP alone does not identify a person.
10:17–10:18	MFA denial, then MFA approval. Student says they approved a prompt around 10:18.	An unexpected prompt may have been approved. Check raw logs for exact ordering.
10:20	Session created from 198.51.100.23; student says no sign-in after 09:05.	Potential unauthorised access is a concern, not a proven conclusion.

The IP addresses are reserved documentation examples. A different IP is a reason to investigate, not proof of an attacker.

C2. Likely explanation, alternative explanation and checks

Likely explanation: Someone may have obtained the student's password and tried to access the account. Repeated password failures and MFA activity may have led to a session after the student approved an unexpected prompt. This is a possibility, not a proven conclusion.

Alternative explanation: Another device, a legitimate background application, or an old session may have generated some activity, and the student may not recognise it.

Two checks: (1) Review raw authentication/session logs, timestamps, MFA records, and device/browser details. (2) Ask whether the student used another device or application and compare recorded details with known devices. An IP address alone cannot identify a person.

C3. Three prioritised response actions

1. Protect the account: revoke suspicious active sessions, reset the password through the official process, and review MFA settings. Tell the student to deny unexpected prompts. 2. Preserve evidence: secure relevant logs, timestamps, session records and MFA events. 3. Investigate and monitor: check unfamiliar devices, account changes and later sign-ins; restore normal access after the account is secured.

C4. Incident summary (60–100 words)

On 18 October 2026, student07 had a password success and MFA approval from the usual classroom IP at 09:00–09:01. At 10:14–10:16, password failures occurred from a different IP. The log then records an MFA denial at 10:17, an MFA approval at 10:18, and a session created at 10:20. The student says they did not sign in after 09:05 but approved a prompt around 10:18. Unauthorised access is suspected, not proven. IT should secure the account, preserve logs, and investigate authentication records.

Task D – Review an AI Security Answer (25 Marks)

D1. Five unsafe or unsupported claims/actions

AI claim/action	Correction
“The email is safe because it uses HTTPS.”	HTTPS encrypts traffic but does not prove a website is genuine. Verify through the official college portal or independently obtained contact.
“The logs prove the person using 198.51.100.23 stole the password.”	Logs show activity from an IP, not a person's identity or how a password was obtained.
“MFA prevents every account compromise.”	MFA adds protection but users can still be tricked into approving unexpected prompts.
“Delete the suspicious logs.”	Preserve logs and timestamps as evidence for investigation.
“Send the student's password to support.”	Never share passwords or OTPs. Secure the account through the official process and contact authorised IT staff.

D2. Better prompt for an AI tool

“Analyse only the fictional cybersecurity evidence supplied in this assessment. Separate observed facts from assumptions and state what cannot be confirmed. Do not identify or accuse a person without sufficient evidence. Do not request passwords, OTPs, identity documents, or other secrets. Recommend safe actions to protect the account and preserve evidence. Support important claims with trustworthy references. Explain what additional evidence would help. Use simple English and clear headings.”

D3. Corrected security advice (100–150 words)

The suspicious email and sign-in activity should be investigated carefully before reaching a final conclusion. HTTPS does not guarantee that an email or website is genuine. The student should verify scholarship information through the college's official portal or an independently obtained contact number. The logs show activity from a different IP address, followed by password and MFA events and a session creation. However, these records do not prove who used the IP address or how the account was accessed. The authorised IT team should secure the account, revoke suspicious sessions, reset the password, review MFA settings, and preserve relevant logs. The team should check device information and authentication records for further evidence. Passwords and OTPs must never be shared with support staff. Conclusions should be based on verified evidence, not assumptions.

D4. AI-use declaration

I used ChatGPT to help organise the assessment, explain cybersecurity concepts in simple English, and prepare a draft. I reviewed the content and will check it against the assessment instructions and references before submission.

References and evidence notes

1. CISA – Secure Our World (basic account and phishing guidance): https://www.cisa.gov/secure-our-world. This supports general advice about account protection and recognising suspicious messages.

2. NIST – Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile (AI 600-1): https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf. This supports checking AI-generated information and avoiding unsupported certainty.

Additional useful reference: NIST Digital Identity Guidelines: https://pages.nist.gov/800-63-4/.